

# **sercon**

## User Manual

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# sercon manual

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Long-form reference for the `pkg/scriptengine` library, the `sercon` CLI, the built-in script API, and the JavaScript runtime surface that `scriptengine` exposes via [goja](#) and [goja\\_nodejs](#). Examples are runnable — save snippets as `.ts` files and execute with `sercon file.ts` unless otherwise noted.

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## 1. Overview

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`sercon` is a Go program that runs TypeScript files. It's built from two parts:

- `pkg/scriptengine` — a library you embed in your own Go code. You register Go-callable bindings, then call `Run` / `RunFile` to execute a `.ts` script that talks to them.
- `cmd/sercon` — a thin CLI built on the library, useful for ad-hoc test scripts and as a working example of every binding kind.

The runtime is pure Go: [goja](#) is the JS engine, [esbuild](#) (used as a Go library) is the TS→JS transpiler, and [goja\\_nodejs](#) provides Promises, `setTimeout`, `console`, and CommonJS `require`. No cgo, no Node.

## 2. Quickstart

---

Install:

```
go install github.com/codedeviate/sercon/cmd/sercon@latest
```

Run one of the bundled examples:

```
sercon examples/scripts/smoke.ts examples/scripts/async.ts
```

...or run them all via `make demo`. `examples/scripts/` ships a runnable `.ts` (or `.tsx`) file per feature area — `hash.ts`, `strings.ts`, `path-and-time.ts`, `default-export.ts`, `tsx-demo.ts`, and the original `smoke.ts` / `async.ts` / `hang.ts` (`hang.ts` is the timeout demo and intentionally exits non-zero).

Generate a declaration file for editor autocomplete:

```
sercon --emit-dts api.d.ts
```

Embed in your own Go program:

```
eng := scriptengine.New(scriptengine.Options{
    Timeout:    5 * time.Second,
    ScriptRoot: "./scripts",
})
eng.Register("upper", strings.ToUpper)
val, err := eng.RunFile(ctx, "scripts/main.ts")
```

## 3. Library API — `pkg/scriptengine`

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**Engine, Options, New**

```
type Options struct {
    Timeout      time.Duration // wall-clock limit per Run; 0 disables
    ScriptRoot   string        // base for require/import resolution
    DisableConsole bool          // turn off the goja_nodejs console module
    Verbose      io.Writer     // diagnostic traces, prefixed "[sercon] "
    ModuleLoader func(candidatePath string) (source string, found bool,
err error)
}

func New(opts Options) *Engine
```

`ModuleLoader`, when non-nil, is consulted for every require/import candidate path **before** the filesystem — the hook for embedders that want to serve modules from somewhere other than disk (an in-memory FS, a network source, an embedded bundle). goja probes several candidates per specifier (`./x`, `./x.ts`, `./x/index.ts`, ...); the engine also tries the bare path plus the usual extension fallbacks (`.ts` / `.tsx` / `.js` / `.mjs` / `.cjs` / `.json`) against the loader, so a loader can match on a plain suffix. Return `(source, true, nil)` to serve the module (transpiled when the path ends in `.ts` / `.tsx`, exactly like a disk read); `("", false, nil)` to fall through to the filesystem; `("", false, err)` to abort resolution.

```
eng := scriptengine.New(scriptengine.Options{
    ModuleLoader: func(path string) (string, bool, error) {
        if strings.HasSuffix(path, "greeting.ts") {
            return `export const hi = (n: string) => "hi " + n;`, true, nil
        }
        return "", false, nil // fall through to disk
    },
})
```

`Engine` is the host of all registered bindings. Construct it once, then call `Run` / `RunFile` as many times as needed — each call gets a *fresh* `*goja.Runtime` so script state never leaks between invocations.

`ScriptRoot` defaults to the working directory at `New` time if left empty. Both `Timeout` and `ScriptRoot` can be overridden on a per-Run basis only by constructing a fresh `Engine`; see [Out-of-scope](#) for the per-Run override on the roadmap.

## Registering bindings

| Function                                                       | Use when...                                                                                                                                                                                                                                   |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>Register(name, value)</code>                             | The binding is a pure function, struct, or primitive that doesn't need access to the runtime.                                                                                                                                                 |
| <code>RegisterNamespace(name, members map[string]any)</code>   | You want to expose <code>name.x</code> , <code>name.y</code> without defining a Go struct.                                                                                                                                                    |
| <code>RegisterConstructor(name, ctor)</code>                   | The binding is a Go function whose return value should be treated as a class in the emitted <code>.d.ts</code> . (Runtime semantics today are identical to <code>Register</code> ; the <code>d.ts</code> emitter is the only differentiator.) |
| <code>RegisterFactory(name, factory func(vm, loop) any)</code> | The binding needs the per-Run <code>*goja.Runtime</code> or <code>*eventloop.EventLoop</code> in scope (typical for Promise-returning I/O).                                                                                                   |

| Function                                             | Use when...                                        |
|------------------------------------------------------|----------------------------------------------------|
| <code>RegisterNamespaceFactory(name, factory)</code> | Same, but the factory returns the full member map. |

Example — synchronous function binding:

```
eng.Register("greet", func(name string) string { return "hi " + name })
```

Example — namespace with a sync member:

```
eng.RegisterNamespace("math2", map[string]any{
    "square": func(n float64) float64 { return n * n },
    "pi":     3.14159,
})
```

Example — async (Promise-returning) binding via `PromisifyAsync`:

```
eng.RegisterFactory("httpGet", func(vm *goja.Runtime, loop
*eventloop.EventLoop) any {
    return scriptengine.PromisifyAsync(vm, loop,
        func(ctx context.Context, call goja.FunctionCall) (map[string]any,
error) {
            url := call.Argument(0).String()
            resp, err := http.Get(url)
            if err != nil {
                return nil, err
            }
            defer resp.Body.Close()
            body, _ := io.ReadAll(resp.Body)
            return map[string]any{"status": resp.StatusCode, "body":
string(body)}, nil
        })
})
```

## Run, RunFile

```
func (e *Engine) Run(ctx context.Context, name, source string, opts
...RunOption) (goja.Value, error)
func (e *Engine) RunFile(ctx context.Context, path string, opts
...RunOption) (goja.Value, error)
```

`Run` executes `source` as an entry-script TS. `name` is used in stack traces. The returned value is currently always `undefined` — top-level expression capture is on the backlog.

Both methods respect `Options.Timeout` and `ctx` cancellation, whichever fires first; the resulting error is either `ErrScriptTimeout`, `ctx.Err()`, or the underlying JS exception.

### Per-Run options

```
type RunOption func(*runConfig)

func WithScriptRoot(dir string) RunOption
```

Reuse a single Engine across many scripts that each live in their own directory by passing `WithScriptRoot(dir)` to `Run` / `RunFile`. The override applies only to this call; `Options.ScriptRoot` is used for every other `Run`.

```
_, _ = eng.Run(ctx, "main.ts", source,
scriptengine.WithScriptRoot("/path/to/run42"))
```

```
func WithArgs(args []string) RunOption
```

Set the per-script argument vector. The script reads it as `runtime.argv`, a global with Node/Bun layout: `argv[0]` is `Options.ProgramName` (defaults to `filepath.Base(os.Args[0])`); the CLI sets it to `"sercon"`, `argv[1]` is the running script's path, and the `WithArgs` values follow from index 2. The global is always present — with no args it is just `[programName, scriptPath]`.

```
_, _ = eng.Run(ctx, "main.ts", source, scriptengine.WithArgs([]string{"--
port", "8080"}))
// inside the script: runtime.argv === [ProgramName, "/abs/main.ts", "--
port", "8080"]
```

### Reset

```
func (e *Engine) Reset()
```

Clears every registered binding. Useful when a long-lived Engine is reused across unrelated batches of scripts that each want a clean global namespace. **Not safe to call concurrently with `Run` / `RunFile`.**

### WriteTypes

```
func (e *Engine) WriteTypes(w io.Writer) error
```

Emits a `.d.ts` describing the registered surface. See section [14. Type generation](#) for what the mapping looks like.

## SetDocs and SetMemberDocs

```
func (e *Engine) SetDocs(path, doc string)
func (e *Engine) SetMemberDocs(namespace string, docs map[string]string)
```

Attach JSDoc strings to registered bindings so the emitted `.d.ts` grows readable editor hover. `SetDocs` takes a dotted path — either a bare top-level name (`"greet"`, `"http"`) or `"namespace.member"` for namespace members (`"http.get"`, `"exec.shell"`); `SetMemberDocs` is the convenience for documenting many members of a namespace at once (the namespace itself can still be documented separately via `SetDocs`).

Multi-line docs are written as `\n`-separated strings; the emitter expands them into a standard `*`-prefixed JSDoc block. Single-line docs collapse to `/** ... */`. Calling `SetDocs` with an empty string removes any previous doc for that path. Bindings without a doc entry get no JSDoc block at all — the emitter doesn't insert empty `/** */` placeholders.

`SetDocs` may be called before or after the matching `Register...` call; docs are looked up by name at emit time. Like the registration methods, `SetDocs` must not race with a `Run` / `WriteTypes` / `Reset` on the same engine.

```
eng.Register("greet", func(name string) string { return "hi " + name })
eng.SetDocs("greet", "Greet someone by name.")

eng.RegisterNamespace("math2", map[string]any{
    "pi":      3.14159,
    "squared": func(n float64) float64 { return n * n },
})
eng.SetDocs("math2", "Tiny math helpers.")
eng.SetMemberDocs("math2", map[string]string{
    "pi":      "Circumference / diameter ratio.",
    "squared": "Multiply a number by itself.",
})
```

## PromisifyAsync[T] and AsyncBinding

```
type AsyncBinding struct {
    Func      func(goja.FunctionCall) goja.Value
    TSReturnType string
}

func PromisifyAsync[T any](vm *goja.Runtime, loop *eventloop.EventLoop,
    work func(ctx context.Context, call goja.FunctionCall) (T, error),
) AsyncBinding
```

`PromisifyAsync` turns blocking Go work into a JS Promise. It launches the work in a goroutine, parks a `SetTimeout` sentinel so `loop.Run` doesn't return early, and schedules the

resolve/reject back onto the event loop. **Required** for any Promise-returning binding — `RunOnLoop` alone is not counted as a live job by the event loop.

`PromisifyAsync` returns an `AsyncBinding` carrier rather than the bare callback. The engine unwraps it to `.Func` at `vm.Set` time so goja's special-case detection of `func(goja.FunctionCall) goja.Value` host-callbacks still fires; the `.d.ts` emitter reads `.TSReturnType` to emit `Promise<T>` (mapped from the generic `T` via `reflect` at construction time) instead of the previous `unknown`. Hosts assigning the result into a `map[string]any` for `RegisterNamespace` / `RegisterNamespaceFactory` get the unwrap recursively too — no manual work required.

## Version

```
import "github.com/codedeviate/sercon/pkg/scriptengine"

fmt.Println(scriptengine.Version) // "0.1.0"
```

Bumped in lockstep with the git tag.

## 4. CLI — `sercon`

```
sercon [flags] <script.ts> [script.ts ...]
sercon [flags] - # read entry script from stdin
sercon --examples | --help | --version
```

| Flag                                  | Purpose                                                                                                            |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <code>-timeout DURATION</code>        | Wall-clock limit per script (default <code>10s</code> ; <code>0</code> disables).                                  |
| <code>-root DIR</code>                | Override the script root for <code>require</code> / <code>import</code> resolution.                                |
| <code>-emit-dts PATH</code>           | Write the example bindings' <code>.d.ts</code> to <code>PATH</code> and exit.                                      |
| <code>-v</code>                       | Verbose: trace the rewritten entry-script JS and each module resolution to stderr; also print duration on failure. |
| <code>-h</code> , <code>--help</code> | In-depth colored help.                                                                                             |
| <code>--examples</code>               | In-depth colored walkthrough of every feature.                                                                     |
| <code>--version</code>                | Print the engine version (plus goja/esbuild build-info versions).                                                  |
| <code>--watch</code>                  | Re-run on file change under the script root. Debounced 150 ms. Ctrl-C exits.                                       |



Each positional argument is either a path to a `.ts` / `.tsx` file or `-` to read an entry script from standard input:

```
echo 'runtime.log(1 + 2);' | sercon -  
sercon prelude.ts -                # prelude then stdin
```

## Passing arguments: `--` and `runtime.argv`

Everything after a standalone `--` is the script argument vector, exposed to every script via `runtime.argv`:

```
sercon run.ts -- --port 8080 hello
```

`runtime.argv` uses the Node/Bun layout `[programName, scriptPath, ...userArgs]` — `argv[0]` is `"sercon"`, `argv[1]` is the path of the script currently executing, and the post-`--` arguments start at index 2 (so `runtime.argv.slice(2)` is just your args). All scripts in one invocation share the same `--` tail, but each sees its own path at `argv[1]`. The global is always present; with no `--` it is just `[programName, scriptPath]`.

```
const args = runtime.argv.slice(2); // ["--port", "8080", "hello"]
```

Exit codes are distinct per failure type so shells can react sensibly. When several scripts run, the highest applicable code wins:

| Code | Meaning                                                                           |
|------|-----------------------------------------------------------------------------------|
| 0    | every script passed.                                                              |
| 1    | CLI usage error (unknown flag, missing script argument, ...).                     |
| 2    | at least one script failed to transpile — never ran.                              |
| 3    | at least one script timed out ( <code>-timeout</code> ) or was context-cancelled. |
| 4    | at least one script ran and threw a JS exception.                                 |

`-v` writes lines prefixed with `[sercon]` to stderr. The traces include the full rewritten entry-script JS (the form goja actually runs, after the ESM→CJS rewrite + async IIFE wrapper) and every module-resolution event, so debugging an unexpected resolve target or a mis-rewritten import is straightforward.

The CLI registers ten reserved top-level globals; see the next section.

## `--watch`: re-run on file change

`sercon --watch <script.ts>` runs the script once and then blocks, re-running it every time a watched file changes under the script root. Useful for iterating on a script body or on the

binding surface during development.

```
sercon --watch examples/scripts/smoke.ts
```

What's watched:

- The script root (resolved the same way as the one-shot mode — `-root DIR` if set, else `dirname` of the first script argument).
- Recursively. Symlinks aren't followed.
- New directories that appear during the session are picked up automatically.

What's filtered:

- **File extensions:** `.ts` / `.tsx` / `.js` / `.jsx` / `.json` trigger a re-run; `.d.ts` files match via suffix too. Anything else (Markdown, images, editor swap files, build artefacts) is ignored.
- **Directories:** hidden directories (`.git`, `.vscode`, anything starting with `.`) and `node_modules` are excluded — both because they generate floods of irrelevant events and because recursing into them inflates the watcher's directory count for no gain.

Re-runs are **debounced 150 ms**. Editors typically fire several events per save (write → rename → chmod); collapsing them into a single re-run keeps the loop responsive without thrashing. The debounce window resets on every event, so a rapid burst of saves becomes one re-run after the burst settles.

Each run is delimited by a line like `--- sercon re-run @ HH:MM:SS ---` so the output is visually distinct from the previous run.

`Ctrl-C` (SIGINT) or `SIGTERM` exit cleanly. Per-script failures inside a watch session log as `FAIL` but don't terminate the loop — a syntax error you're trying to fix shouldn't kick you out of watch mode.

The watcher reuses the same `Engine` across runs. Each `Run` already gets a fresh `*goja.Runtime`, so re-running is just re-invoking `runOne` on the existing engine — there's no per-iteration setup cost beyond the transpile + module-resolution work.

**Module-graph invalidation.** Watch mode tracks each entry script's import graph (its own file plus every module it resolves, captured via `Engine.SetResolveHook`) during the run. On a file change it re-runs **only the entries whose graph includes the changed file** — editing a helper that just one of three entry scripts imports re-runs that one entry, not all three. An entry's own file counts (editing it re-runs it); stdin entries and any entry whose graph isn't known yet re-run unconditionally (conservative). Before each re-run the module cache is busted (`Engine.ResetModuleCache`) so an edited import's new source actually takes effect — the registry otherwise caches compiled bytecode across runs.

**`sercon serve`: long-running scripts with production niceties**

```
sercon serve [flags] <script.ts> [-- args...]
```

`sercon serve` is a sibling subcommand that wraps the same engine but adds the conveniences a process supervisor (systemd, docker, launchd) usually wants from a long-lived script. Use it whenever a script binds an HTTP/HTTPS listener (or any future `server.*` protocol) and is expected to keep running. Vanilla `sercon script.ts` also keeps the loop alive while listeners are bound, but gives you none of the deltas below.

| Flag                                     | Purpose                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--shutdown-timeout DURATION</code> | Graceful-shutdown deadline on SIGTERM/SIGINT (default <code>30s</code> ). After the deadline the engine hard-cancels.                                                                                                                                                                                                         |
| <code>--port-override N</code>           | If non-zero, replaces every literal <code>port:</code> in <code>server.*.listen({...})</code> calls with <code>N</code> . Useful for spinning up the same script on a different port without editing source. Only literal port values are rewritten — computed expressions ( <code>{port: getPort()}</code> ) are left alone. |
| <code>-timeout DURATION</code>           | Per-script wall-clock limit. Defaults to <code>0</code> (disabled) under <code>serve</code> so listeners don't get cut off; opt back in if you want one.                                                                                                                                                                      |
| <code>-root DIR</code>                   | Script root for <code>require / import</code> resolution (same semantics as vanilla <code>sercon</code> ).                                                                                                                                                                                                                    |
| <code>-v</code>                          | Verbose engine tracing to stderr.                                                                                                                                                                                                                                                                                             |

Behavioural deltas vs vanilla `sercon` :

- **Access log to stderr.** Each request emits one line in the format `ts remote method path status dur_μs`, e.g. `2026-05-29T10:23:14Z 127.0.0.1:54321 GET /users/42 200 1843μs`. WebSocket upgrades log only the upgrade line; per-frame traffic is suppressed.
- **Readiness signal to stdout.** When each listener calls back into the binding with `bound`, the binding prints `READY listening on tcp/0.0.0.0:8080` to stdout. Multiple listeners produce one `READY` line each. Supervisors that read-line on startup ( `systemd-notify`, `docker-compose healthchecks` ) can wait on it.
- **Graceful shutdown.** `SIGTERM / SIGINT` triggers `srv.close()` on every active listener concurrently, then waits up to `--shutdown-timeout` for the script to drain. Clean exit is exit code `0`. Past the deadline, `loop.Terminate()` fires and the process exits with the usual classification.
- **No `--watch`.** Re-running while a listener is bound would race port rebinding; `sercon serve --watch ...` exits with a usage error. Use vanilla `sercon --watch script.ts` for the hot-reload development loop.

```
sercon serve examples/scripts/server-http.ts
sercon serve --port-override 9090 server.ts
sercon serve --shutdown-timeout 10s server.ts
```

## 5. Reserved globals (script surface)

sercon scripts get ten reserved top-level globals. Use them directly — there is no enclosing namespace. The full per-binding reference is the generated `examples/scripts/api.d.ts`; this section is the at-a-glance prose.

**Reserved names:** `runtime`, `crypto`, `text`, `codec`, `fs`, `net`, `db`, `server`, `services`, `tui`. User code can shadow these with a local `let / const / var` per normal JavaScript scoping — sercon does not intervene at runtime. `server` (added in v0.10.0) covers inbound listeners — see [section 6](#) for the long-form treatment.

### Migration from v0.8.0

| v0.8.0                                 | v0.9.0                            |
|----------------------------------------|-----------------------------------|
| <code>api.runtime.log</code>           | <code>runtime.log</code>          |
| <code>api.crypto.hash.sha256</code>    | <code>crypto.hash.sha256</code>   |
| <code>api.text.str.trim</code>         | <code>text.str.trim</code>        |
| <code>api.format.barcode.encode</code> | <code>codec.barcode.encode</code> |
| <code>api.fs.path.basename</code>      | <code>fs.path.basename</code>     |
| <code>api.net.http.get</code>          | <code>net.http.get</code>         |
| <code>api.db.sqlite.open</code>        | <code>db.sqlite.open</code>       |
| <code>api.tools.exec.shell</code>      | <code>services.exec.shell</code>  |
| <code>api.tools.git.commit</code>      | <code>services.git.commit</code>  |
| <code>api.ui.tui.layout</code>         | <code>tui.layout</code>           |
| <code>Sercon.argv</code>               | <code>runtime.argv</code>         |

The `api` global no longer exists; reading it throws a `ReferenceError`. The `Sercon` global is also gone.

A mechanical `sed` pass over your scripts handles the prefix drop and the three renames — see the `CHANGELOG.md` [0.9.0] "Changed" entry for a sketch.

#### `runtime`

Script-host scaffolding. Members:

- `runtime.log(...args)` — stringify each argument and `println` the result.
- `runtime.assert.equal(actual, expected, msg?)` / `runtime.assert.ok(cond, msg?)` — throw on mismatch / falsy.
- `runtime.time.nowMs()` — current wall-clock in milliseconds.
- `runtime.time.sleep(ms)` — Promise that resolves after `ms` on the event loop.
- `runtime.time.format(unixMs, layout, tz?)` — Go-style time layout (e.g. "2006-01-02 15:04:05").
- `runtime.env.get(name)` — process environment variable; returns `undefined` when unset (never throws).
- `runtime.argv` — `[programName, scriptPath, ...userArgs]`. User args come from the CLI args after a `--` separator. The layout mirrors Node/Bun: `argv[0]` is "sercon", `argv[1]` is the path of the currently-executing script (so each script in a multi-arg invocation sees its own path), and `argv.slice(2)` is just your args. The property is always present; with no `--` it is just `[programName, scriptPath]`.

```
runtime.log("hello");
runtime.assert.equal(2 + 2, 4);
const args = runtime.argv.slice(2); // post-`--` user args
```

## crypto

Hashing, JWT, and age-style encryption. Members:

- `crypto.hash.*` — `md5`, `sha1`, `sha256`, `sha384`, `sha512`, `sha3_256`, `sha3_512`, `blake3`, `crc32`. All take a `string` and return the hex digest.
- `crypto.jwt.sign(claims, key, opts?)` / `crypto.jwt.view(token)` / `crypto.jwt.validate(token, key, opts?)` — HS256/RS256/ES256 JWTs.
- `crypto.encrypt.{keygen, keygenPgp, encrypt, decrypt, rekey, detectBackend}` — age-format symmetric encryption with PGP-compat key wrapping.

## text

String, regex, charset, and data manipulation. Members:

- `text.str.*` — `trim`, `ltrim`, `rtrim`, `reverse`, `stripHtml`, `nl2br`, `br2nl`, `base64Encode`, `base64Decode`, `urlEncode`, `urlDecode`, `htmlEntityDecode`, `pad` / `lpad` / `rpadd`, `sprintf`, `printf`, `normalizeNewlines`.
- `text.preg.*` and `text.preg2.*` — PCRE-style regex (PHP-compatible named-capture, lookbehind, etc.).
- `text.charset.detect(data)` / `decode(data, charset)` / `encode(text, charset)` — character-set detection and conversion.
- `text.jq.query(json, expr)` / `queryAll(json, expr)` — jq filtering against parsed JSON values.
- `text.diff.compare(a, b, opts?)` — unified / patience / inline diffs between two strings.

## codec

Binary-format codecs (was `format` in v0.8.0). Members:

- `codec.compression.{algorithms, compress, decompress}` — gzip, deflate, zlib, bzip2, zstd, brotli, lz4, xz, snappy.
- `codec.barcode.{formats, decodableFormats, encode, decode}` — QR, DataMatrix, Aztec, PDF417, Code128, Code39, Codabar, EAN-13/8, UPC-A, ITF. `encode` returns PNG bytes; `decode` reads image bytes.
- `codec.checkdigit.{algorithms, validate, compute, inspect}` — Luhn, ISBN-10/13, EAN-13/8, UPC-A.

## fs

Filesystem operations:

- `fs.path.dirname(p) / fs.path.basename(p, suffix?)` — POSIX-style.
- `fs.archive.create(srcPath, opts?) / fs.archive.extract(archivePath, destDir, opts?)` — tar/zip with optional compression.

## net

Network clients and probes:

- `net.http.{get, post, request}` — fetch-style HTTP client; `request` accepts headers, body, timeout, retry, follow, and basic auth.
- `net.probe.{tcp, dns, tls, ntp, whois, ping, smtp, wss}` — one-shot reachability / handshake probes. Each returns a structured result; failures surface as `Error` objects with details.
- `net.netstatus.check(host)` — combined reachability summary.
- `net.email.*` — `spf`, `dmARC`, `mtaSTS`, `tlsRPT`, `bIMI`, and `all(domain)` which runs every probe in parallel and aggregates; plus `send({...})` — outbound SMTP sender (see §6.7).
- `net.browser.open(url)` — launches the OS default browser.

## db

Database / KV / directory clients:

- `db.sqlite.open(path)` — embedded SQLite (cgo-free driver).
- `db.redis.open(addr, opts?)` — Redis client.
- `db.memcached.open(addr)` — Memcached client.
- `db.ldap.open(url, opts?)` — LDAP client (search, bind).
- `db.dict.{define, match}` — local dictionary lookup / fuzzy match.

## services

Subprocess and external-CLI / service wrappers (was `tools` in v0.8.0). Members:

- `services.exec.shell(cmd, opts?)` — run a shell command; captures stdout/stderr or streams into a `tui` pane when `opts.pane` is set.
- `services.exec.http(method, url, opts?)` — shell-level `curl`-style HTTP (separate from `net.http`, intended for raw protocol use).
- `services.git.*` — git porcelain wrappers (clone, status, commit, log, ...) invoking the local git binary.
- `services.gh.{authStatus, prList, repoView}` — thin `gh` CLI wrappers.
- `services.ai.{providers, send}` — provider-agnostic AI chat client.

## tui

Multi-pane terminal UI (was `ui.tui` in v0.8.0). `tui.layout(tree)` declares panes;

`tui.pane(name)` returns a pane handle with `write`, `writeln`, `clear`, `title`.

`services.exec.shell({pane})` streams subprocess I/O into a pane.

Scripts that orchestrate multiple subprocesses (e.g. `brew`, `npm`, `cargo` updates running in parallel) can declare a multi-pane terminal layout and route each subprocess's stdout/stderr into its own pane. Activation is **script-driven**: the first call to `tui.layout(...)` enters TUI mode. If stdout is not a TTY (CI, pipes, `make demo`), the same calls fall back to prefixed plain-text lines ( `[paneName] line` ) so the same script runs in both contexts.

## Layout

```
tui.layout({
  rows: [
    { name: "log", title: "Orchestrator", weight: 1 },
    { cols: [
      { name: "brew" },
      { name: "npm" },
    ],
      weight: 2,
    },
  ],
});
```

Each layout node is one of:

- `{ name: string, title?: string, weight?: number }` — a leaf pane.
- `{ rows: LayoutNode[], weight?: number }` — a vertical split.
- `{ cols: LayoutNode[], weight?: number }` — a horizontal split.

`weight` defaults to 1. Pane names must be unique across the whole tree. `layout` may be called **once** per Run; a second call throws.

## Pane handles

```
const log = tui.pane("log");
log.writeln("Updating Homebrew...");
log.title("Done");
log.clear();
log.write("partial line ");
```

`tui.pane(name)` returns a handle with `write`, `writeln`, `clear`, `title`. Methods are synchronous from the script's perspective — they enqueue to the TUI goroutine.

## Subprocess routing

`services.exec.shell(cmd, opts)` accepts `opts.pane` (a Pane handle or a pane name string):

```
await services.exec.shell("brew update && brew upgrade", { pane: "brew" });
```

When set, the subprocess's stdout **and** stderr stream into the pane line by line. The returned `{ exitCode, durationMs, success }` is the same as without `pane`: except `stdout` and `stderr` are empty (data was streamed, not captured). ANSI colors are translated to tview color tags and rendered natively; `\r` without `\n` overwrites the current line so progress spinners render cleanly.

## Keybindings (TTY mode only)

| Key             | Action                       |
|-----------------|------------------------------|
| Tab / Shift-Tab | Cycle focus through panes    |
| PgUp / PgDn     | Scroll focused pane one page |
| ↑ / ↓           | Scroll focused pane one line |
| Home / End      | Jump to top / bottom         |
| Ctrl-C          | Abort the script             |

The focused pane has a yellow border; the bottom status bar shows its name and the active keys.

## Limitations (v1)

- `tui` is **incompatible with** `--watch`: calling `tui.layout()` under `--watch` throws. Use one or the other.
- No mouse, no runtime layout mutation, no input panes (`tui.input`), no snapshot-to-normal-screen on exit. The alt screen is restored at script end and the usual `PASS / FAIL` line prints.



## 6. Servers

---

The `server` global (added in v0.10.0) hosts inbound listeners — scripts that *accept* connections rather than make them. The first sub-spec ships `server.http` and `server.https`; future cycles will add SMTP, IMAP, FTP, and POP3 against the same engine foundation.

### 6.1 Foundation: `HoldRun` + `LoopCallable`

Two engine primitives back every long-lived binding. They live on `*scriptengine.Engine` (CLI use of them is internal; the public contract is "use these if you write a similar binding yourself").

**`Engine.HoldRun(reason string) (release func())`** — keeps `loop.Run` from exiting while a binding has resources outstanding. The event loop normally exits as soon as its `jobCount` drops to zero; `setTimeout` / `setInterval` / `setImmediate` are the only things that count. `HoldRun` parks a 24-hour sentinel timer under the hood (same trick `PromisifyAsync` uses for one-shot async work), returns a `release` function that clears it, and increments a refcount so multiple concurrent holders compose. `release` is idempotent. Any sentinels still parked when `Run` returns are cleanup-drained automatically as a safety net — a binding that forgets to call `release` does not leak into the next `Run`.

**`scriptengine.NewLoopCallable(loop, fn)`** — wraps a captured `goja.Callable` (a JS function reference) so it can be invoked from *any* goroutine. Goja's runtime is single-threaded; calling a `Callable` directly from a non-loop goroutine corrupts engine state.

`LoopCallable.Call(buildArgs)` enqueues a `loop.RunOnLoop` callback, builds the arg list on the loop (where `vm.ToValue` is valid), invokes the callable, and returns the result to the caller's goroutine. `LoopCallable.CallOnLoop(vm, args...)` is the already-on-the-loop variant — using `Call` from inside a loop callback deadlocks because the new `RunOnLoop` job can't run until the current one returns.

You use them together. A typical server binding holds one or two `LoopCallable`s for the user's handlers, parks one `HoldRun` sentinel per active listener, and releases on close. The CLI bindings under `cmd/sercon/api_server*.go` follow this pattern.

**Concurrency model.** Because every handler invocation marshals back onto the single goja loop, **handlers serialize**: only one JS handler runs at a time, across all listeners. This is a feature (no JS data races, no shared-state confusion) and a throughput ceiling. For CPU-bound work in a handler, offload to a Go binding that does the work in a goroutine and resolves a Promise.

### 6.2 `server.http.listen` / `server.https.listen`

```
// HTTP
const srv = await server.http.listen({
  port: 8080, // required
  host: "0.0.0.0", // default
  routes: { "GET /": (req, res) => res.text("hi") },
  use: [logger], // optional global middleware
});

// HTTPS – identical plus cert/key (file paths OR inline PEM strings)
const srv2 = await server.https.listen({
  port: 8443,
  cert: "/etc/ssl/server.pem",
  key: "/etc/ssl/server.key",
  routes,
});

srv.address; // "tcp/0.0.0.0:8080"
await srv.close(); // graceful: stop accepting, drain, release HoldRun
await srv.stopped; // Promise that resolves when the listener exits
```

### Options:

| Key    | Type                       | Notes                                                                                            |
|--------|----------------------------|--------------------------------------------------------------------------------------------------|
| port   | number                     | Required. Bind port. 0 lets the kernel pick; read the chosen port off <code>srv.address</code> . |
| host   | string                     | Default "0.0.0.0". Pass "127.0.0.1" for loopback-only.                                           |
| routes | Record<string, RouteValue> | Required (may be <code>{}</code> ). Keys are stdlib <code>http.ServeMux</code> patterns.         |
| use    | Middleware[]               | Optional global middleware; runs in array order before any per-route middleware.                 |
| cert   | string                     | <b>HTTPS only.</b> File path or inline PEM.                                                      |
| key    | string                     | <b>HTTPS only.</b> File path or inline PEM.                                                      |

**Route patterns** are stdlib Go 1.22+ `http.ServeMux` syntax: "METHOD /path/{param}/{rest...}" . The leading method is optional (omit to match any). `{name}` captures one path segment; `{name...}` captures the tail (wildcard). No new routing library — stdlib does the matching, including longest-prefix wins. Captured values land in `req.params` .

**RouteValue** is either a bare handler — `(req, res) => res.text("...")` — or an object `{ use?: Middleware[], handler: HandlerFn }` for per-route middleware (see §6.4).

**Request shape** (`req`):

```

type Request = {
  method:    string;           // "GET"
  url:       string;           // full URL incl. scheme + host
  path:      string;           // "/users/42"
  query:     Record<string, string[]>; // ?a=1&a=2 → {a: ["1","2"]}
  headers:   Record<string, string[]>; // lowercase keys
  params:    Record<string, string>;   // path-pattern captures
  body:      string;           // UTF-8 decoded request body
  bodyBytes: Uint8Array;       // same data as raw bytes
  remote:    string;           // "1.2.3.4:54321"
  cookies:   Record<string, string>;   // parsed Cookie header
};

```

**Response builder** ( `res` ) — every method returns `res` for chaining; the terminal `.json` / `.text` / `.html` / `.bytes` / `.empty` / `.redirect` sets the body and flips an internal `finalized` flag:

```

type CookieOpts = {
  domain?:  string;
  path?:    string;
  maxAge?:  number;           // seconds; 0 = session; <0 = delete
  expires?: number;           // unix ms
  secure?:  boolean;
  httpOnly?: boolean;
  sameSite?: "strict" | "lax" | "none";
};

type Response = {
  status(code: number): Response;
  header(name: string, value: string): Response;
  cookie(name: string, value: string, opts?: CookieOpts): Response;
  // Terminals (all return res so they remain chainable but no further
  // write makes sense after one fires; a second terminal throws):
  json(value: unknown): Response; // → application/json
  text(s: string): Response;      // → text/plain
  html(s: string): Response;      // → text/html
  bytes(b: Uint8Array, ct?: string): Response; // default
  application/octet-stream
  empty(): Response;              // no body; pair with .status() for
  204/304
  redirect(loc: string, code?: number): Response; // default 302
  // Upgrade (WebSocket):
  upgradeWebSocket(opts?: { readBuffer?: number }): Promise<WebSocket>;
};

```

After the handler's returned Promise resolves, the engine checks `res.finalized`:

- **true** — buffered status / headers / body get written to the underlying `http.ResponseWriter`.
- **false** — engine sends 204 No Content.
- **handler threw** (sync or async) — engine sends 500 Internal Server Error and logs the error (`stderr` under vanilla `sercon`; access log under `sercon serve`). The script keeps running; only that one request is affected.

Calling a terminal twice on the same `res` throws a `TypeError` (the second call sees `finalized === true` and refuses).

**Multiple listeners** compose naturally. A script can run

```
const [a, b] = await Promise.all([
  server.http.listen({ port: 80, routes }),
  server.https.listen({ port: 443, routes, cert, key }),
]);
```

Each `listen` call holds its own `HoldRun` sentinel; closing one listener does not affect the other.

## 6.3 Static-file serving

```
"GET /assets/{rest...}": server.http.static({
  dir:          "/var/www/public",    // root directory on disk
  stripPrefix:  "/assets/",          // URL prefix to strip before disk
  lookup
  index?:      "index.html",         // accepted but currently unused
  (stdlib default applies)
  etag?:       true,                 // accepted but currently unused
  (stdlib default applies)
}),
```

`server.http.static({...})` returns an opaque handler the route compiler unwraps and registers directly on the mux. **The route pattern must include a `{rest...}` wildcard** — without it the match only ever produces the literal prefix and no subpath ever resolves on disk. Internally a `stdlib http.FileServer(http.Dir(dir))` wrapped in `http.StripPrefix(stripPrefix, ...)`; ETag and range requests work out of the box. Symlinks outside `dir` are blocked (`stdlib http.Dir`'s default). No directory listing customisation yet — `index` and `etag` options are reserved for future tuning; v0.10.0 inherits the `stdlib` defaults.

## 6.4 Middleware

Onion model. A middleware is `(req, res, next) => Promise<void>`. `next` is an async function that runs the rest of the chain; awaiting it lets the middleware post-process.

```
const logger = async (req, res, next) => {
  const start = runtime.time.nowMs();
  await next(); // run downstream chain
  runtime.log(req.method, req.path, "->", runtime.time.nowMs() - start,
"ms");
};

await server.http.listen({
  port: 8080,
  use: [logger], // global middleware
  routes: {
    "GET /api/secure": { // per-route middleware
      use: [authCheck],
      handler: (req, res) => res.json({ ok: true }),
    },
  },
});
```

#### Attachment points:

- **Global** — `use`: array in `listen({...})` options. Runs for every request, in declaration order, before any per-route middleware.
- **Per-route** — when a route value is `{ use: [...], handler: fn }`, those middleware run after globals but before the handler.

For a request to `POST /api/secure` with global middleware `[A, B]` and per-route middleware `[C]`, the composition is:

```
A(req, res, () => B(req, res, () => C(req, res, () => handler(req, res))))
```

Each layer can `await next()` then mutate response headers or record timings, exactly like Express / Koa.

**Short-circuit:** a middleware that does *not* call `await next()` stops the chain. It must terminate `res` itself (e.g. `res.status(401).text("denied")`) — otherwise the engine sends `204 No Content` per the unfinalized-response rule.

**Throwing:** any throw mid-chain (sync or via rejected Promise) bubbles to the engine's 500 path. Wrap with `try / catch` if you want custom error handling.

## 6.5 WebSocket upgrade

```
"GET /ws": async (req, res) => {
  const ws = await res.upgradeWebSocket({ readBuffer: 64 });
  // ws is both an async iterator AND has .send / .close.
  for await (const msg of ws) {
    if (msg.type === "text") {
      await ws.send("echo:" + msg.text);
    } else {
      await ws.send(msg.bytes);          // msg.type === "binary"
    }
  }
  // iterator ends on close; ws.closeCode / ws.closeReason populated
},
```

### Type:

```
type WSMMessage =
  | { type: "text";   text: string   }
  | { type: "binary"; bytes: Uint8Array };

type WebSocket = AsyncIterable<WSMessage> & {
  send(data: string | Uint8Array): Promise<void>;
  close(code?: number, reason?: string): Promise<void>;
  closeCode?: number;
  closeReason?: string;
  remote: string;
};
```

**Library:** [github.com/coder/websocket](https://github.com/coder/websocket) (the modern successor to [nhooyr.io/websocket](https://nhooyr.io/websocket); [gorilla/websocket](https://gorilla.github.io/websocket) is in maintenance mode). Pure Go, zero deps.

**Marshalling.** Per upgraded connection, a Go goroutine reads frames into a buffered channel (capacity = `readBuffer`, default 64). Each `next()` on the async iterator pulls one message off the channel and resolves on the loop via `LoopCallable`. `ws.send()` schedules a write back through the connection-owning goroutine.

**Backpressure.** If the read buffer fills (slow JS consumer), back-pressure propagates into the websocket library's read loop — the client eventually sees `1009 message too big` after the library's own limit (1MB default). Bump `readBuffer` if your workload bursts and you want more in-flight queueing.

`ws.close()` closes the send channel, drains the receive channel, sends a close frame, and ends the async iterator on the next `next()` call. The script's `for await` exits cleanly; the handler's Promise resolves; the connection's goroutine returns. The HTTP server's `HoldRun` is **unaffected** — it stays alive for new connections until `srv.close()`.

`async function*` and `for await (...)` are not natively parsed by goja. esbuild's `Supported` flag lowers both during transpile (see CHANGELOG), and every Run installs

`Symbol.asyncIterator = Symbol.for("@@asyncIterator")` so the lowered helper and user code agree on the same iteration key.

## 6.6 Lifecycle

**Vanilla sercon script.ts.** Each `server.*.listen` call parks a `HoldRun` sentinel and the event loop stays alive while the listener is bound. `SIGINT` terminates abruptly: the engine's interrupt watcher calls `loop.Terminate()`, the cleanup drain runs (closing each listener), and the process exits. No access log, no readiness signal, no shutdown timeout.

**sercon serve script.ts.** Adds the production niceties documented in §4: structured access log to stderr, a `READY` listening on `tcp/...` line on stdout per listener, and graceful shutdown with `--shutdown-timeout` (default `30s`). Clean `SIGTERM` exits `0`. Choose `serve` whenever the script is supervised (systemd, docker compose, launchd).

```
# Quick local test:
sercon examples/scripts/server-http.ts

# Production-style supervised run:
sercon serve examples/scripts/server-http.ts
```

## 6.7 SMTP

SMTP is a **stateful, multi-stage transaction**: a client connects, optionally authenticates, declares a sender (`MAIL FROM`), one or more recipients (`RCPT TO`), then streams the message body (`DATA`). `sercon` maps each stage to a JavaScript callback so a script can accept or reject the transaction incrementally — refuse an unknown sender at `MAIL`, refuse an unroutable recipient at `RCPT`, or inspect the parsed message at `DATA`. The inbound listener is `server.smtp.listen({...})`; the outbound side is `net.email.send({...})` (covered at the end of this section).

**`server.smtp.listen({...})`**

Synchronously binds the listening socket (so a port already in use throws immediately), then serves in the background. Returns a handle once the loop schedules it.

```

const srv = await server.smtp.listen({
  port: 2525,
  hostname: "mx.example.com",
  handlers: {
    onMail: (env)      => env.from.endsWith("@example.com"),
    onRcpt: (env, rcpt) => isLocalMailbox(rcpt),
    onData: (env, msg) => { store(msg); return true; },
  },
});

// srv.address  → "tcp/0.0.0.0:2525"
// srv.close()  → Promise<void> (graceful, 30s drain)
// srv.stopped  → Promise<void> (resolves when the listener exits)

```

### Options:

| Field             | Type                                            | Default          | Notes                                                  |
|-------------------|-------------------------------------------------|------------------|--------------------------------------------------------|
| port              | number                                          | —                | <b>Required.</b> TCP port to bind.                     |
| host              | string                                          | "0.0.0.0"        | Bind address.                                          |
| hostname          | string                                          | os.Hostname()    | EHLO greeting + Received: identity.                    |
| handlers          | {onMail, onRcpt, onData}                        | —                | <b>Required.</b> All three functions; see below.       |
| auth              | (user, pass, env)<br>=> bool  <br>Promise<bool> | —                | Optional SASL verifier (PLAIN + LOGIN).                |
| starttls          | {cert, key}                                     | —                | Enables STARTTLS. File paths <b>or</b> inline PEM.     |
| allowInsecureAuth | boolean                                         | false            | Permit AUTH on a plaintext connection (dev only).      |
| maxMessageBytes   | number                                          | 10485760 (10 MB) | DATA size cap; a non-positive value keeps the default. |
| maxRecipients     | number                                          | 100              | Max RCPT TO per transaction.                           |



| Field                       | Type                | Default            | Notes                                   |
|-----------------------------|---------------------|--------------------|-----------------------------------------|
| <code>sessionTimeout</code> | <code>number</code> | <code>30000</code> | Per-session idle timeout, milliseconds. |

**Handlers.** `onMail(envelope)`, `onRcpt(envelope, recipient)`, and `onData(envelope, message)` are invoked at the matching protocol stage. All three are required. Each may be `async` — a returned Promise is awaited before the SMTP response is written. The return value (or a thrown exception) controls the SMTP reply, uniformly across all three:

| Return value                  | SMTP response                      | Use case                              |
|-------------------------------|------------------------------------|---------------------------------------|
| <code>true / undefined</code> | <code>250 OK</code>                | Accept the stage                      |
| <code>false</code>            | <code>550 Command refused</code>   | Generic permanent reject              |
| <code>string</code>           | <code>550 &lt;string&gt;</code>    | Permanent reject with a reason        |
| <code>throw</code>            | <code>451 Temporary failure</code> | Transient error (client should retry) |

**Envelope** (passed to all three handlers; `recipients` grows as `RCPT TO` accumulates):

```
type Envelope = {
  from:          string;    // "" for the null sender (DSN bounces)
  recipients:    string[];  // accepted RCPT TO so far
  remote:        string;    // "1.2.3.4:54321"
  helo:          string;    // EHLO/HELO hostname the client gave
  authenticatedUser?: string; // set if AUTH succeeded
  tls?:          { version: string; cipher: string }; // set under
STARTTLS
};
```

**Message** (only `onData`; parsed via `jhillierd/enmime`, with the exact original bytes always available as `raw`):

```

type Message = {
  from:    string;           // From: header (may differ from
  envelope.from)
  to:      string[];         // To: header
  cc:      string[];         // Cc: header
  subject: string;
  headers: Record<string, string[]>; // lowercase keys; multi-valued
  body: {
    text: string;           // text/plain part; "" if absent
    html: string;           // text/html part; "" if absent
  };
  attachments: Array<{
    filename:    string;
    contentType: string;
    bytes:       Uint8Array;
  }>;
  raw: Uint8Array;           // exact DATA bytes (post dot-
  unstuffing)
};

```

Forwarders and DKIM-style signature verifiers should work from `raw`; everyone else uses the parsed fields. `enmime` is lenient with malformed messages, so a script needing strict RFC 5322 conformance can re-parse `raw` itself.

**AUTH.** Supply an `auth` callback to enable SASL. Both **PLAIN** and **LOGIN** mechanisms are advertised; the callback receives the decoded (`username`, `password`, `envelope`) and returns a boolean (or a Promise of one). By default AUTH is refused on a plaintext connection — the client must complete STARTTLS first. Set `allowInsecureAuth: true` to accept credentials over plaintext; this is a **dev-only** escape hatch for trusted local networks and should never be set in production.

**STARTTLS.** Pass `starttls: {cert, key}` (file paths or inline PEM, same convention as `server.https`) to advertise STARTTLS. The upgrade happens mid-session on the same connection; once active, `envelope.tls` is populated. The listener itself is plaintext at bind time — there is no implicit-TLS (SMTPS / port 465) inbound mode.

**Concurrency.** Each connection runs on its own Go goroutine, but every handler invocation **serializes on the goja loop** — exactly one handler runs at a time (the same single-threaded model as the HTTP listener in §6.2). A slow `onData` therefore blocks other connections' handlers. For slow work, acknowledge the stage (`return true`) and process in the background (start a Promise without awaiting it).

A short, self-contained round-trip (bind, send to self, assert receipt):

```

const port = 38095;
let captured: { subject: string; from: string } | null = null;

const srv = await server.smtp.listen({
  port,
  hostname: "test.local",
  handlers: {
    onMail: () => true,
    onRcpt: () => true,
    onData: (env, msg) => {
      captured = { subject: msg.subject, from: env.from };
      return true;
    },
  },
});

await net.email.send({
  to:      "alice@test.local",
  from:    "bob@test.local",
  subject: "round-trip demo",
  body:    "hello from the SMTP demo",
  server:  { host: "127.0.0.1", port, tls: "none" },
});

runtime.assert.equal(captured!.subject, "round-trip demo", "subject");
await srv.close();

```

### **net.email.send({...}) — outbound**

The outbound counterpart lives under `net.email` (next to the `spf` / `dmARC` / ... probes). It opens **one TCP connection per call** — no pooling, no queueing, no automatic retries — composes the MIME body in-tree, and returns a per-recipient outcome.

```

const result = await net.email.send({
  to:      ["alice@example.com", "bob@example.com"], // string OR string[]
  from:    "noreply@my.app",
  subject: "your verification code",
  body:    "your code is 123456",                    // plain text
  html:    "<p>your code is <b>123456</b></p>",        // optional HTML
  alternative
    attachments: [
      { filename: "qr.png", contentType: "image/png", bytes: qrBytes },
    ],
    headers: { "X-App": "myapp" },                    // optional extra
  headers
    server: {
      host: "smtp.example.com",
      port: 587,                                       // default 587
    (submission)
      auth: { username: "u", password: "p" },         // optional PLAIN
    auth
      tls: "starttls",                               // "starttls"
    (default) | "tls" | "none"
    },
    timeout: 30000,                                   // ms; default 30s
  });

// result = { accepted: string[], rejected: Array<{ address, reason }> }

```

**MIME layout.** text only → text/plain; charset=utf-8 (no multipart). text + html → multipart/alternative. Any attachments → multipart/mixed (each part base64, Content-Disposition: attachment). Date, Message-ID, and MIME-Version are auto-generated; headers are merged on top.

**TLS modes.** "starttls" (default) connects plaintext then upgrades, refusing AUTH until TLS is active. "tls" is implicit TLS from connect (SMTPS, typically port 465). "none" is plaintext throughout, with AUTH disabled.

**Outcomes.** Transport-level failures (connection refused, TLS handshake, AUTH, MAIL FROM rejected) **throw** a JS exception. The rejected array captures only individual RCPT TO addresses the server permitted us to attempt but then refused — the transaction continues for the rest, so a partial send still delivers to the accepted recipients.

## 7. JavaScript runtime built-ins (goja)

scriptengine runs on goja, which implements **ES5.1 + a large subset of ES6+**. The following are present and behave per spec:

## Globals

- `Object`, `Array`, `String`, `Number`, `Boolean`, `BigInt`
- `Math`, `Date`, `JSON`, `RegExp`
- `Error`, `TypeError`, `RangeError`, `SyntaxError`, `ReferenceError`, `EvalError`, `URIError`
- `Promise` (provided via goja's native implementation; resolution microtasks are pumped by the event loop)
- `Symbol` (partial: well-known symbols `Symbol.iterator`, `Symbol.asyncIterator`, etc., are supported)
- `Proxy`, `Reflect` (partial)

```
runtime.log(Math.PI.toFixed(4), Math.max(1, 9, 3));
runtime.log(new Date().toISOString());
runtime.log(JSON.stringify({ a: 1, b: [2, 3] }));
runtime.log("abc".repeat(3), "abc".padStart(6, "_"));
```

## Collections

- `Map`, `Set`, `WeakMap`, `WeakSet`

```
const counts = new Map<string, number>();
for (const ch of "banana") counts.set(ch, (counts.get(ch) ?? 0) + 1);
runtime.log([...counts]); // [{"b",1}, {"a",3}, {"n",2}]
```

## Typed arrays

- `ArrayBuffer`, `DataView`
- `Int8Array`, `Uint8Array`, `Uint8ClampedArray`, `Int16Array`, `Uint16Array`, `Int32Array`, `Uint32Array`, `Float32Array`, `Float64Array`

```
const buf = new ArrayBuffer(4);
new DataView(buf).setUint32(0, 0xCAFEBADE);
runtime.log(new Uint8Array(buf)[0].toString(16)); // ca
```

## Iteration and generators

- `for...of`, `for...in`, `spread`, `destructuring`
- Generator functions (`function*`)
- Async generators (`async function*`) — supported via goja's event loop

```
function* fib() {
  let [a, b] = [0, 1];
  while (true) {
    yield a;
    [a, b] = [b, a + b];
  }
}
const it = fib();
const first10 = Array.from({ length: 10 }, () => it.next().value);
runtime.log(first10);
```

## Not (yet) provided

- `fetch` — use `net.http.*` from the CLI, or register your own binding.
- `Worker`, threading primitives.
- DOM globals (`window`, `document`).
- Native ES modules (`import` / `export` are *transpiled* to CommonJS at load time — see [section 9](#)).

## 8. Async runtime additions (goja\_nodejs)

The event loop bundles a small Node-compatible runtime on top of goja:

### Timers

```
setTimeout(() => runtime.log("after 50ms"), 50);
setInterval(() => runtime.log("tick"), 100);
setImmediate(() => runtime.log("next tick"));
clearTimeout(t); clearInterval(i); clearImmediate(im);
```

The event loop holds the script alive while *any* timer is pending. If your script ends with only `setTimeout` callbacks scheduled, the run will wait for them to fire before returning.

### console

```
console.log("info");
console.error("oops");
console.warn("careful");
console.info("fyi");
console.debug("trace");
```

Disable with `Options.DisableConsole = true` if you want a clean global namespace.

## `require`

CommonJS `require`, with TS-aware extension fallback added by `sercon` (see [section 11](#)):

```
const helpers = require("./helpers");
helpers.doThing();
```

Both `require("./foo")` and `import { x } from "./foo"` resolve to the same module instance per Run; esbuild rewrites `import` to `require` at transpile time.

## 9. TypeScript support

TS is transpiled by esbuild in-process. There is no `tsc`, no `tsconfig.json`, no type-checking — types are erased and the resulting JS is what executes. Practical implications:

- All TS syntax esbuild supports is allowed: type annotations, `interface`, `type`, generics, enums (compiled to objects), namespaces, decorators (legacy & TC39 stage 3).
- `import type { X } from "y"` is stripped entirely (no runtime require).
- TS errors are *not* surfaced as build errors — only esbuild parse errors are. Run a separate `tsc --noEmit` in CI if you want type enforcement.
- `tsconfig.json` is **not** consulted.

`.tsx` and `.jsx` are supported via esbuild's `LoaderTSX` / `LoaderJSX`, including for required modules resolved by the engine's extension fallback. JSX has no React runtime in scope by default — either set the factory at the file level with an `@jsx` pragma:

```
/** @jsx h */
function h(tag: string, props: any, ...children: any[]) {
  return { tag, props: props ?? {}, children };
}
export const Box = (label: string) => <div className="box">{label}</div>;
```

...or provide a `React.createElement`-compatible binding from Go and rely on esbuild's default pragma. ESM `export default <value>` also works through the entry-script rewriter's `__esModule ? .default : m` unwrap, so `import answer from "./mod"` resolves to the default export.

## 10. Top-level `await`

Top-level `await` works in entry scripts:

```
const r = await net.http.get("https://example.com");
runtime.log(r.status);
```

How it works under the hood: esbuild rejects top-level await with the `cjs` output format, so the engine transpiles the *entry* script with `format=esm`, then rewrites `import` statements to `require()` calls and wraps the rest of the body in:

```
;(async () => { /* your transpiled body */ })().then(__resolve, __reject);
```

`__resolve` and `__reject` are bindings the engine sets on the runtime to capture the final settlement. *Required-module* code (anything loaded through `require`) is transpiled with `format=cjs` and does **not** support top-level await — wrap in `(async () => ... )()` yourself if you need it inside a module.

## 11. Module resolution

The engine plugs a custom source loader into `goja_nodejs/require` that adds:

1. **Direct path:** if the requested file exists on disk, use it. `.ts` / `.tsx` files are transpiled on the fly.
2. **Extension fallback** when the request has no extension: try `.ts`, `.tsx`, `.js`, `.cjs`, `.mjs`, `.json` in that order.
3. **.js → .ts swap:** if a path ending in `.js` / `.cjs` / `.mjs` is requested but the literal file doesn't exist, the same path ending in `.ts` (or `.tsx`) is tried. Handles `package.json` `main` fields that point at compiled output where only the TypeScript source is on disk.
4. **package.json source preference:** when reading a `package.json`, if a `source` field is present and points at an existing `.ts` / `.tsx` file, `main` is rewritten to that path before the registry consumes it. Matches the convention used by parcel, microbundle, and similar bundlers to mark the TS source of truth.
5. **Directory index:** if the resolved path is a directory, try `index.ts`, `index.tsx`, `index.js`.

The base directory follows Node's CommonJS rules — relative imports resolve against the requiring module's directory, courtesy of `goja_nodejs`.

```
// All resolve to ./helpers/assert.ts:
import { check } from "./helpers/assert";
import { check } from "./helpers/assert.ts";
const { check } = require("./helpers/assert");
const { check } = require("./helpers/assert.js");
```

JSON imports work out of the box via either the ESM-style default import (esbuild rewrites it to a `require`) or a direct `require`:

```
import data from "./data.json";
const r = require("./data.json");
```



`node_modules` lookup *works* via the upstream registry, but the source loader doesn't currently transpile through that path — keep TS helpers under `ScriptRoot`.

## 12. Timeouts and cancellation

---

Two mechanisms interrupt a running script:

- `Options.Timeout` — wall-clock limit per `Run`. Exceeding it returns `scriptengine.ErrScriptTimeout`.
- `ctx` passed to `Run / RunFile` — cancelling the context returns `ctx.Err()` (typically `context.Canceled` or `context.DeadlineExceeded`).

Both call `vm.Interrupt(...)` and `loop.Terminate()` from a watcher goroutine. This works for **sync** JS — including `while(true){}` — because the goja VM checks the interrupt flag between bytecode steps. A host Go binding that's blocked in cgo or a long syscall isn't interruptible mid-call; if you write such a binding, plumb the engine `ctx` through to it yourself.

```
eng := scriptengine.New(scriptengine.Options{Timeout: 200 *
time.Millisecond})
_, err := eng.Run(ctx, "loop.ts", `while (true) {}`)
// err is scriptengine.ErrScriptTimeout, returned ~200-300ms after Run.
```

## 13. Error semantics

---

- A Go binding returning `(T, error)` automatically throws as a JS exception when `err != nil`. The exception's `.message` is `err.Error()`.

```
try { mayFail(); } catch (e) { runtime.log(String(e)); }
```

- A Go binding that `panic`s with a `*goja.Object` (e.g. `vm.NewGoError(...)`) does the same.
- A script throwing a JS error out of the top level surfaces from `Run` as a Go `error` whose message is the JS error's `.message`.
- Timeout errors satisfy `errors.Is(err, scriptengine.ErrScriptTimeout)`. Cancellation errors satisfy `errors.Is(err, context.Canceled)` or `context.DeadlineExceeded`.

## 14. Type generation (.d.ts)

---

`Engine.WriteTypes(w)` walks the registered bindings and emits a TypeScript declaration file. The mapping table (abbreviated):

| Go type                                                                             | TS type                                                    |
|-------------------------------------------------------------------------------------|------------------------------------------------------------|
| <code>string</code>                                                                 | <code>string</code>                                        |
| <code>any numeric / float64</code>                                                  | <code>number</code>                                        |
| <code>bool</code>                                                                   | <code>boolean</code>                                       |
| <code>[]T</code>                                                                    | <code>T[]; []byte → Uint8Array</code>                      |
| <code>map[string]T</code>                                                           | <code>Record&lt;string, T&gt;</code>                       |
| <code>*T</code>                                                                     | <code>T</code> (transparent unwrap)                        |
| <code>error</code> (single return)                                                  | omitted (collapses to <code>void</code> )                  |
| <code>(T, error)</code> (tuple return)                                              | <code>T</code>                                             |
| <code>interface{}</code> / <code>any</code>                                         | <code>unknown</code>                                       |
| <code>goja.FunctionCall</code> parameter                                            | folded into <code>(...args: unknown[])</code>              |
| <code>goja.Value</code> return                                                      | <code>unknown</code>                                       |
| <code>scriptengine.AsyncBinding</code> value (from <code>PromisifyAsync[T]</code> ) | <code>Promise&lt;T&gt;</code>                              |
| self-referential types                                                              | <code>unknown</code> (cycle detection bails after depth 4) |

```
sercon --emit-dts api.d.ts
```

In your editor, place the resulting `.d.ts` next to your scripts (or reference it via `/// <reference path="..." />`) for autocomplete.

## JSDoc comments on declarations

The emitter pulls JSDoc strings from the engine's doc map (set via `Engine.SetDocs` / `Engine.SetMemberDocs`) and renders them above each declaration. Single-line docs collapse to `/** ... */`; multi-line docs expand to a standard `*`-prefixed block. Bindings without a doc entry emit no JSDoc — the emitter never inserts empty placeholders. Docs are looked up by the same dotted path the binding was registered under, so namespace members get hover text too:

```
// AUTO-GENERATED by scriptengine.Engine.WriteTypes – do not edit by hand.

/** Hashing primitives. */
declare const crypto: {
  hash: {
    /** SHA-256 hex digest of a UTF-8 input. */
    sha256(...args: unknown[]): string;
    /** BLAKE3 hex digest (32-byte output, lukechampine.com/blake3). */
    blake3(...args: unknown[]): string;
    // ...
  };
  // ...
};
```

The CLI's reserved-global surface ships with a curated doc map; see `cmd/sercon/api_docs.go` for the source of truth and add new entries there when adding new bindings (the lockstep rule keeps `--examples` / `MANUAL` / `api.d.ts` in sync, and the doc map is now the seventh artifact in that chain).

## 15. Limitations and gotchas

- **No native ES modules at runtime.** All `import` is rewritten to `require` by esbuild. Side-effect-only imports run the module body exactly once per Run.
- **No top-level await inside required modules.** Wrap with `(async () => ... )()` inside the module if you need it.
- **No tsconfig.json honoured.** Module resolution and type checking are independent from any project-level TS config.
- **Per-Run runtime.** Side effects on `globalThis` do not persist between calls to `Run` on the same `Engine`. The `require` *compile* cache is shared; module *exports* are not.
- **Multi-line ESM imports may stress the entry rewriter.** The current rewriter is line-based and handles the common forms; especially gnarly inputs (comments inside the spec list, very unusual whitespace) fall back to executing as-is, which may throw.
- **RegisterConstructor is d.ts-only today.** At runtime it behaves like `Register`. True `new`-able constructor semantics are on the roadmap.
- **HTTP bindings are real network calls.** `net.http.*` uses `net/http` with a 5s timeout. They are not mockable from JS.

See [OUT-OF-SCOPE.md](#) for the active backlog of deferred ideas.

---

*This manual covers sercon v0.11.2. Whenever you add, remove, or change a flag, a binding, or the script API, update this file alongside the help screen (`--help`), the examples walkthrough (`--examples`), and the `CHANGELOG.md`.*